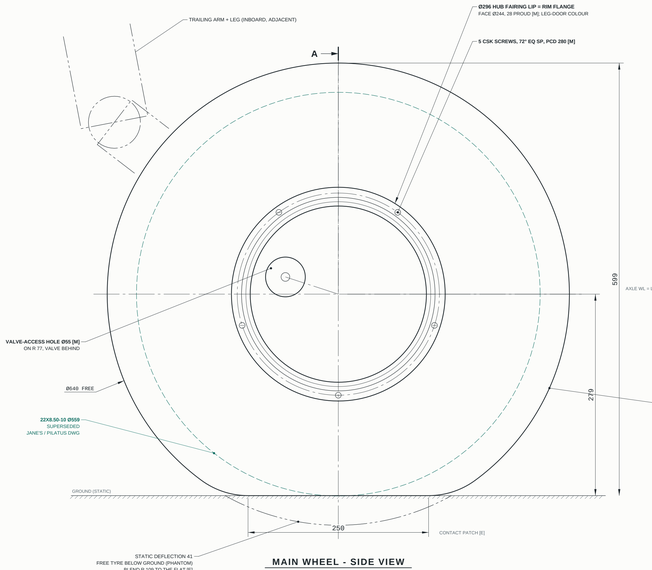
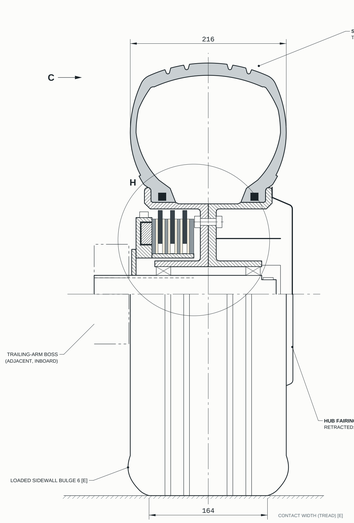


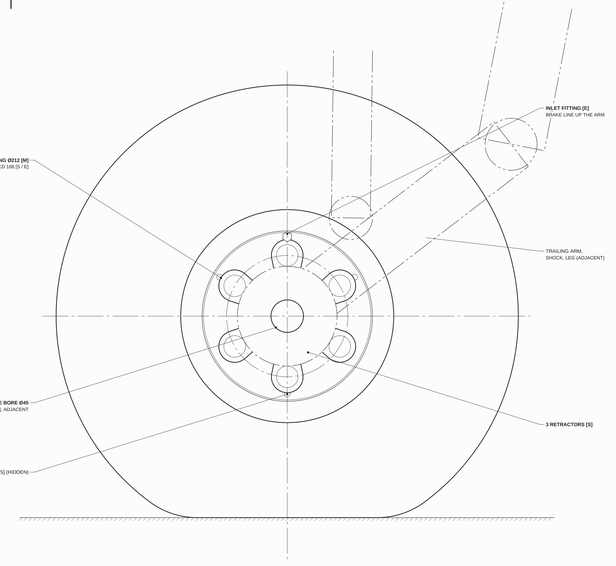
MAIN WHEEL - SIDE VIEW

PORT UNIT SEEN FROM PORT (OUTBOARD FACE) · 1:3



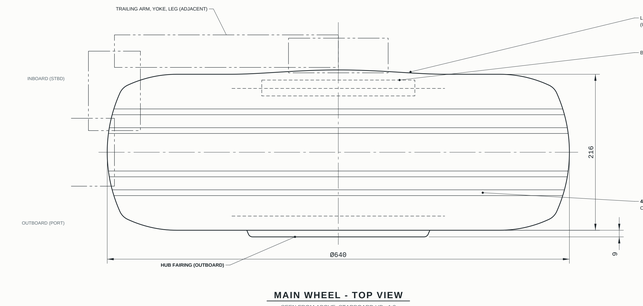
SECTION A-A (HALF)

FRONT VIEW FROM AHEAD, UPPER HALF CUT THROUGH THE AXLE · 1:3



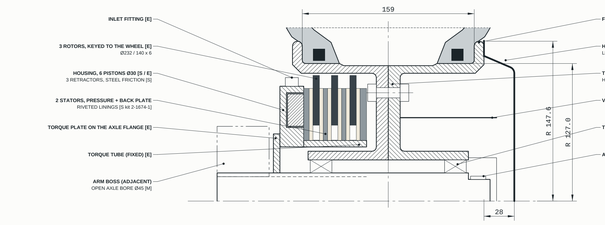
VIEW C - MAIN WHEEL INBOARD FACE

SEEN FROM STARBOARD (BRAKE SIDE), REAR VIEW · 1:3



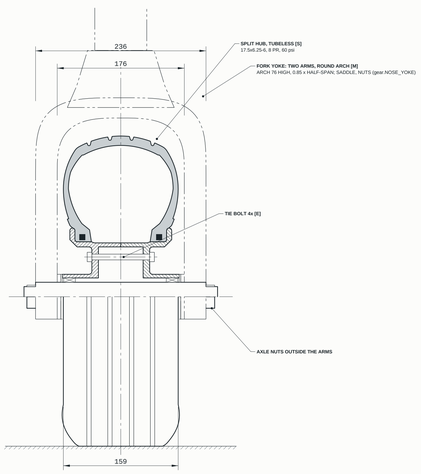
MAIN WHEEL - TOP VIEW

SEEN FROM ABOVE, STARBOARD UP · 1:3



DETAIL H - MAIN HUB AND BRAKE

HALF SECTION A-A, SCALE 1:2

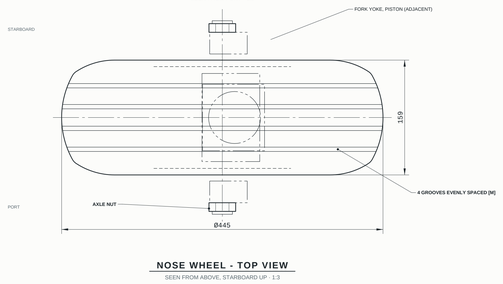


NOSE WHEEL - SIDE VIEW

SEEN FROM PORT · 1:3

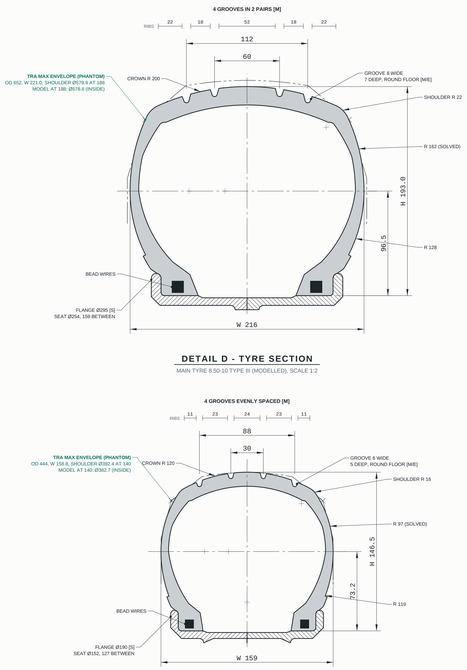
SECTION B-B (HALF)

FRONT VIEW FROM AHEAD, UPPER HALF CUT · 1:3



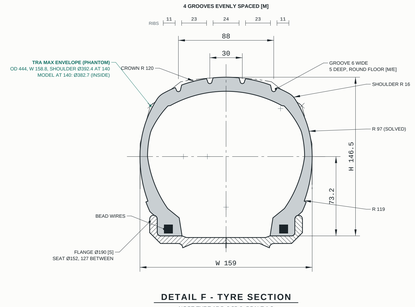
NOSE WHEEL - TOP VIEW

SEEN FROM ABOVE, STARBOARD UP · 1:3



DETAIL D - TYRE SECTION

MAIN TYRE 8.50-10 TYPE III (MODELLED), SCALE 1:2



DETAIL F - TYRE SECTION

NOSE TYRE 17.5x6.25-6, SCALE 1:2

KEY DIMENSIONS - model/wheels.py TABLES

ITEM (mm)	MAIN	NOSE	SOURCE
Tyre (modelled)	8.50-10 Type III	17.5x6.25-6	O owner decision 2026-09-26; S tyre makers / parts lists
Part numbers	8507B6-3 / 025-350-0	175K8881 / 021-327-0	S Goodyear / Michelin (parts listings)
Ply / type / pressure	10 PR TL, 60 psi	8 PR TL, 60 psi	S data book; P0H placard, GSG 02527
Free OD / section W	640 / 216	445 / 159	S TRA 627-652 / 208-221; M photos 620-650
Axle WL = loaded R / defl.	279 / 41	207 / 15.5	G main axle kept; M main 270-280, nose 200-210
Contact patch / blend R	250 / 109	130 / 78	E (free chords 313 / 163)
Section H / aspect	193.0 / 0.89	146.5 / 0.92	D (TRA 0.90 / 0.92)
Bead seat / flange dia	254 / 295.1	152 / 190	S TRA rim 10 / 6 in, flange .81 / .75 in
Between flanges / ledge	158.8 / 34.3	127 / 22.9	S TRA 6.25 / 5.00 in; 1.35 / 0.90 in
Crown R / tread width	200 / 164	120 / 116	E main, M nose / M
Shoulder R / sidewall R	22 / 162	16 / 97	E / D (solved tangent)
Grooves: centres +-	30, 56 (2 pairs)	15, 44 (even)	M head-on photos 3036, 3008, 3001
Groove width x depth	8 x 7	6 x 5	M width / E depth
Ribs centre / mid / outer	52 / 18 / 22	24 / 23 / 11	D outer rib to the shoulder
Wheel	split hub, 2 halves	split hub, 2 halves	S P0H 7-4-11; Goodrich 3-1543 / 3-1501
Tie bolts n on PCD	8 on 200	4 on 110	E hidden (main); M PCD, E count (nose)
Web face s (dish depth)	11 (split plane)	40 (21 deep)	E main (covered); M nose (shallow dish, bolts seen)
Valve / safety plug	R 77 / 1	R 66 / 1	M position; S P0H
Fusible plugs	3 (inboard half)	-	S P0H 7-4-11; E position
Hub fairing lip / face / proud	296 / 244 / 28	-	M 3036, 3005, 3019, 3008, NGX
Fairing screws / hole	5 on 280 / Ø55 @R77	-	M 5 at 72 deg
Brake	6 pist., 3 retr., steel	-	S P0H 7-4-10; kit 2-1674-1
Brake discs	3 rotors / 2 stators	-	S stators / E rotors
Housing lobes / rotor dia	212 / 232	-	M lobe tips / E
Brake stack s	-100 , -20	-	E (s < 0 inboard)
Axle dia / bore	52 / 45	40	G gear.py / M open bore
Fork arms (faces) / boss	-	+-88.118 / 83	M two arms (head-on photos)
Fork arch height / arm fore-aft	-	76 / 42-52	M 3001 head-on, 3036 mx4
Rim-protector step above flange	17 x 2.5	13 x 2.0	M position / E height
Flange-face bore chamfer	8	10	M flange ring width

O owner decision · P proposed · S sourced · M photo-measured · E estimated · D derived · G gear.py. s < 0 = main inboard (brake) side.

PHOTO CHECK - DEVIATION CALL-OUTS (mm)

ITEM	PHOTOS [M]	TABLE / MODEL	Δ	ACTION
Main tyre OD (modelled)	620-650	640	0	owner decision 2026-09-26
22x8.50-10 (superseded)	-	559	-61	MAIN_TYRE_CHOICE
3-D gear.MAIN_TYRE	-	640	0	from wheels.py
Main section width	214-224	216	0	-
Main loaded radius	270-280	axle 279	0	41 flat drawn
Tyre R / fairing lip R	2,15-2.21	2.16	0	-
Retracted tyre depth	~25 (P0H 1 in)	23.5	-1.9	20-30 (fit_check 5)
Main grooves / W	0.14, 0.26	0.14, 0.26	0	3-D geometry
Render groove shader	-	off (geometry)	-	done
Nose tyre OD	17.5 in	445	0	-
Nose loaded radius	200-210	axle 207	0	15.5 flat drawn
Nose grooves / W	0.09, 0.28	0.09, 0.28	0	-
Nose fork arms	2	gear.py 2	0	3-D yoke, round arch, saddle
Nose yoke arch height / half-span	0.85 (3001)	0.86 (p 3.1)	0	round arch (r2: square portal)
Nose arm width fore-aft	42-48	42-52	0	r2: 50-92 (saddle)
Nose axle nut / lock plate	dark hex + tear-drop	3-D both arms	0	steel_dark
Nose tie-bolt PCD	~0.58 flange R	110	0	3036 nose-hub zoom
Hub fairing dia / proud	290-300 / 25-30	296 / 28	0	face 9 > W/2
Main pressure	60 psi placard	gear.py info 60	0	-
Main brake	6-lobe, bright cast	3-D: brake_housing	0	-
Main inboard tones	lobes > flange	housing 0.80 / wheel 0.24	0	r2: one grey
Loaded tread on the ground	flat across W	flat (3-D)	0	r2: crown rib only

MATERIALS (model\assemble.py NAMES)

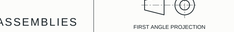
PART	MAIN	NOSE	NOTE
Tyre sidewall / tread / grooves	tire / _tread / _groove	same	matte dusty / satin / dark; NO lettering
Wheel halves	wheel_main	wheel	weathered cast alu (main) / gloss white (nose)
Hub fairing	hub_fairing	-	leg-door blue (3008), a shade darker, soft coat
Brake housing / discs, keys	brake_housing / brake_disc	-	bright satin cast alu / heat-dark steel
Bolts, valve stem	cadmium	cadmium	yellow-chromate cadmium / brass
Fairing screws, valve cap	metal / black	- / black	rubber-sealed cap
Axle nut, hub cap	steel	steel_dark	dark steel nut + tear-drop lock plate (nose)
Bearing caps (nose hub ends)	-	chrome	polished seal rings
Arm / fork	gear_leg	gear_leg	adjacent

LEGEND

- tyre (rubber) in section
- metal in section (hatched)
- brake rotor / stator, lining
- adjacent parts, free tyre, TRA env.
- hidden
- centre line, pitch circle
- 22x8.50-10 tyre (superseded)

SCALE 1:3

SCALE 1:2



NOTES

1. Drawn from the parameter tables of model/wheels.py (PTable: value + source tag + note): tyre envelopes MAIN_TYRE_ENV / NOSE_TYRE_ENV, sections, rims, fairing, brake, axles; never from the mesh. Axle positions and the gear context (trailing arm, leg, shock, nose fork) from model/gear.py, drawn as adjacent parts (phantom). Photos and their measurement working stay in the git-ignored refs/cache.
2. First-angle projection. Each assembly: SIDE view from port (principal), FRONT view from ahead to its right (upper half in section through the axle), TOP view from above below it; for the main wheel also view C from starboard (inboard / brake face) at the far right. Each group's side / front views stand on one ground line.
3. MAIN TYRE - modelled 8.50-10 Type III, OWNER DECISION 2026-09-26 (tyre makers, parts lists, four photo methods; TRA OD 627-652, drawn free OD 640); Jane's / Pilatus drawing 22x8.50-10 (OD 559), superseded, shown in the accent colour. Axles and static stance kept: axle WL 279 = the loaded radius, the tyre flat on the ground over a 250 mm patch and across the tread width (41 mm deflection) with a 6 mm sidewall bulge. Nose tyre loaded too: axle WL 207 (photos 200-210), 130 mm patch, 15.5 mm deflection.
4. 3-D model: gear.MAIN_TYRE = this envelope (R 320.0); the leg door's tyre blister (10 clear, model judging r1) and the round well (bays.WELL_R) follow it. model/wheels.py main_wheel / nose_wheel revolve / extrude the profiles of this sheet (tyre with the grooves, wheel halves, hub fairing, brake, valve, tie bolts); test/consistency_2d3d.py (L-4W) compares the mesh with them.
5. Tyre section: tangent arcs - crown R to the tread half-width, shoulder round, upper sidewall (radius solved: tangent to the shoulder, max width W/2 at 0.50 H), lower sidewall with the moulded rim-protector step (17 / 13 above the flange tip) to where the tyre leaves the rim-flange tip round. At the TRA max shoulder width the modelled section is Ø578.6, inside the TRA max Ø578.9.
6. Main hub fairing (P0H: fairings on the outer hubs), painted in the leg-door colour; its face stands 9 mm outboard of the tyre's max width; retracted (wheel tilted with the skin) it lies 16.0 mm below the skin, inside the tyre's 23.5 mm (P0H -1 in, fit_check 5).
7. Main brake: six pistons, three retractors, steel friction surfaces, bolted to the axle (P0H 7-4-10); 3 rotors / 2 stators inferred from the overhaul kit (24 pads, 2 stators, 48 rivets). Housing, stack and torque tube sized inside the inboard wheel half [E]; one piston lobe is shown in section.
8. No markings (owner decision): no sidewall lettering, brand names or placards on wheels or tyres. The tread grooves are geometry (the render-time groove shader is off). Screw, valve and tie-bolt clocking is the drawn pose (the wheels turn); starboard units are mirror images. The main leg door (LD-1) is not shown.

PILATUS PC-12 PRO

WHEELS & TYRES - MAIN AND NOSE WHEEL ASSEMBLIES

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L4W	A	1:3 / 1:2	A0 · 1 OF 1	mm

DRAWN: Claude - loftkit parametric model

DATE: 2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION